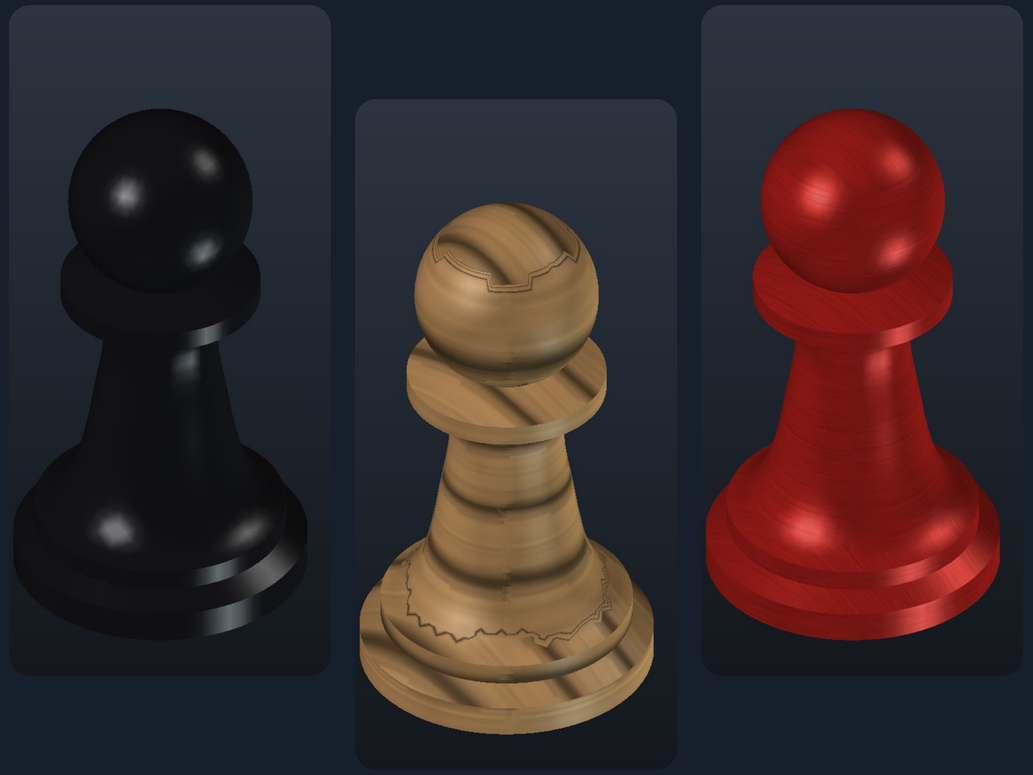
NEW IN THIS BUILD

What's new

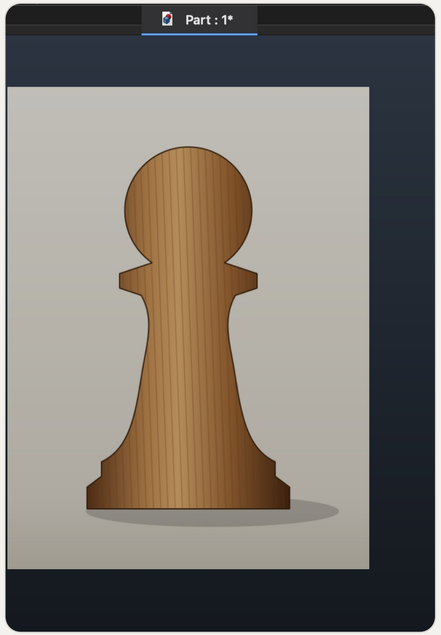
Assembly section and interference, colour and textures, a scaled photo to sketch over, a 3D line, loft planes — and many small things



InnCad Plus 0.9.0 · InnCad 0.13.0 · InCad 0.9.0 (with all of 0.8) · 23 October 2026 · FreeCAD 1.1

DEMO: A CHESS PAWN

From a photo to a part in five steps



1 A photo of a pawn

Image ▸ Insert Image, then On XZ — the picture stands upright



2 Scale

Two points — foot and top, Real length 50 mm; a magnifier under the pointer, 35 % transparency



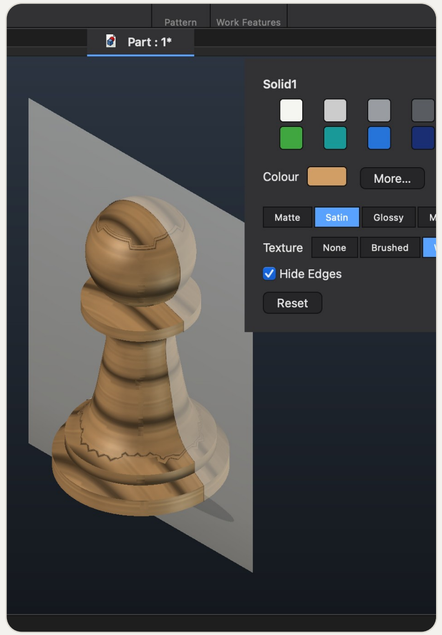
3 Sketch the outline

Half the silhouette over the photo on the XZ plane



4 Revolve

About the Z axis — the pawn is done, exactly 50 mm



5 Appearance

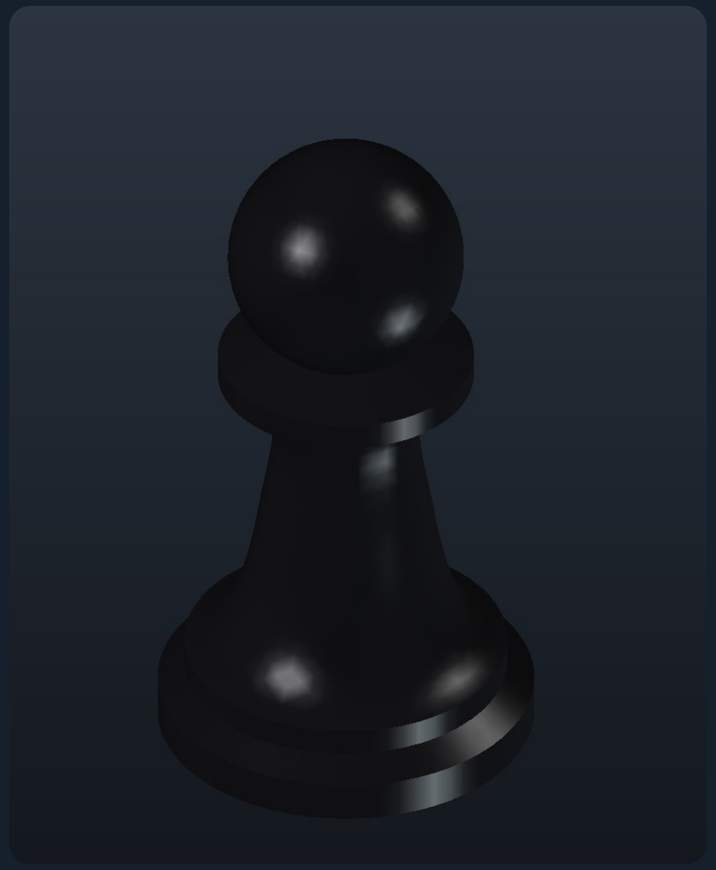
Wood, Satin, Hide Edges — shown on the part at once

APPEARANCE

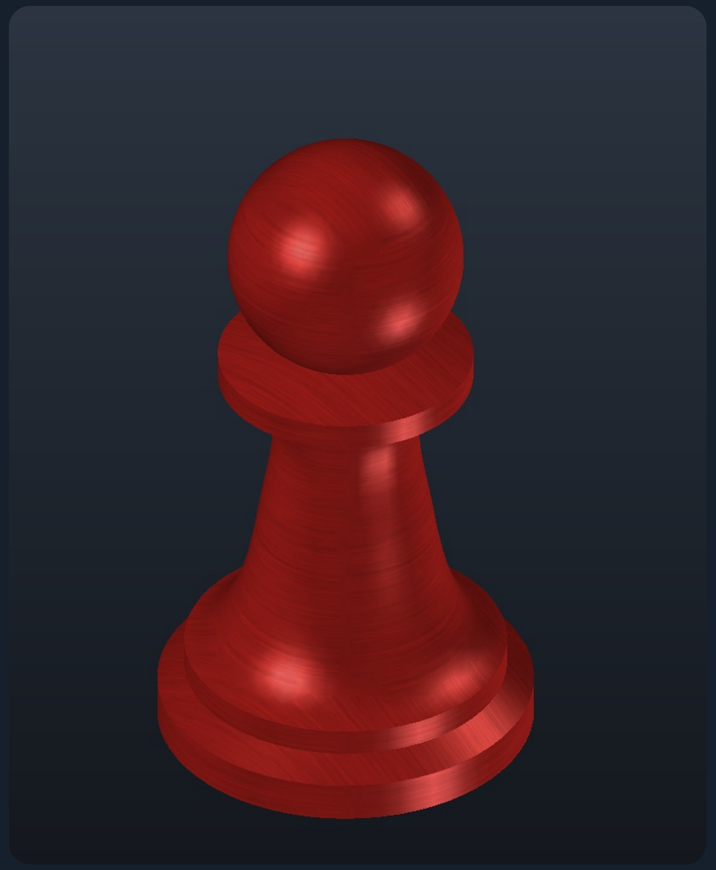
One part, three characters



Wood · Satin



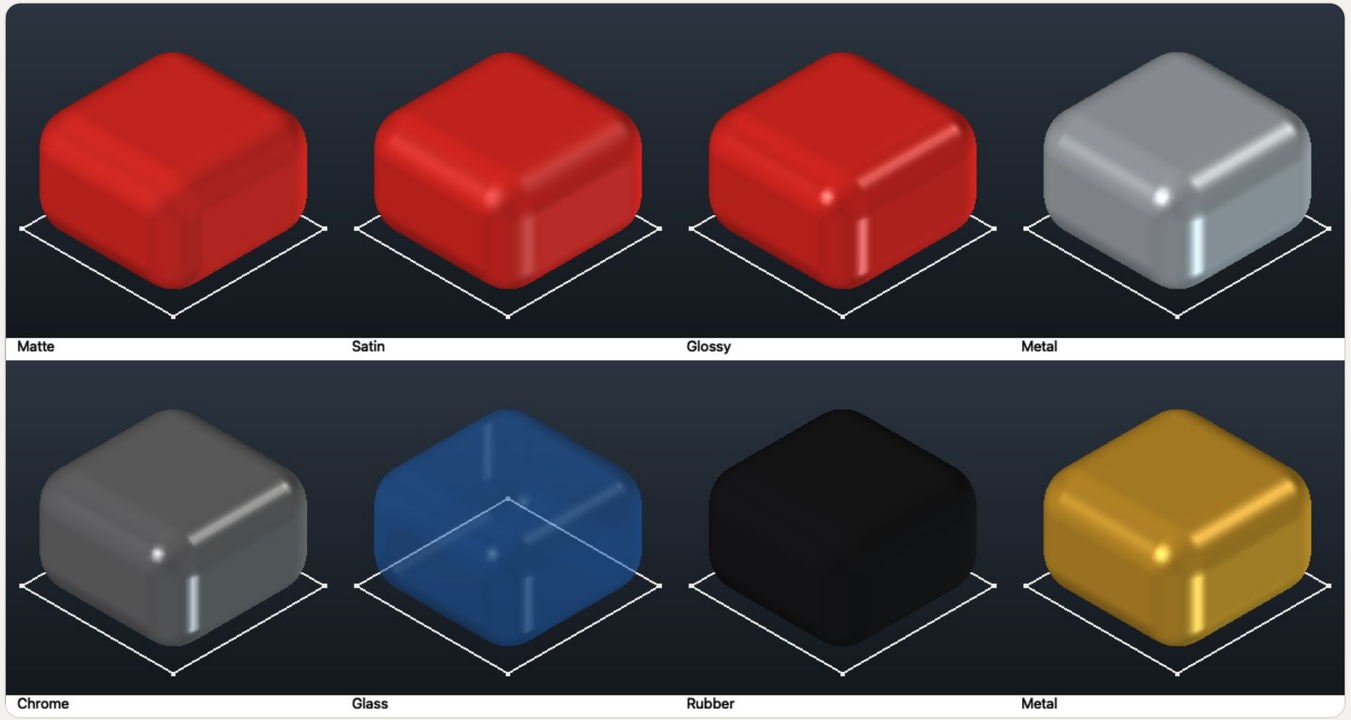
Glossy, black

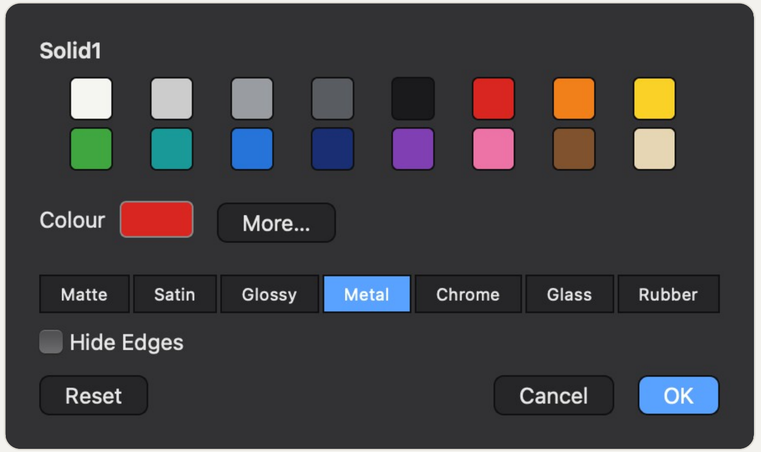


Metal + Brushed

TOOLS ▸ MATERIAL ▸ APPEARANCE

Colour and finish

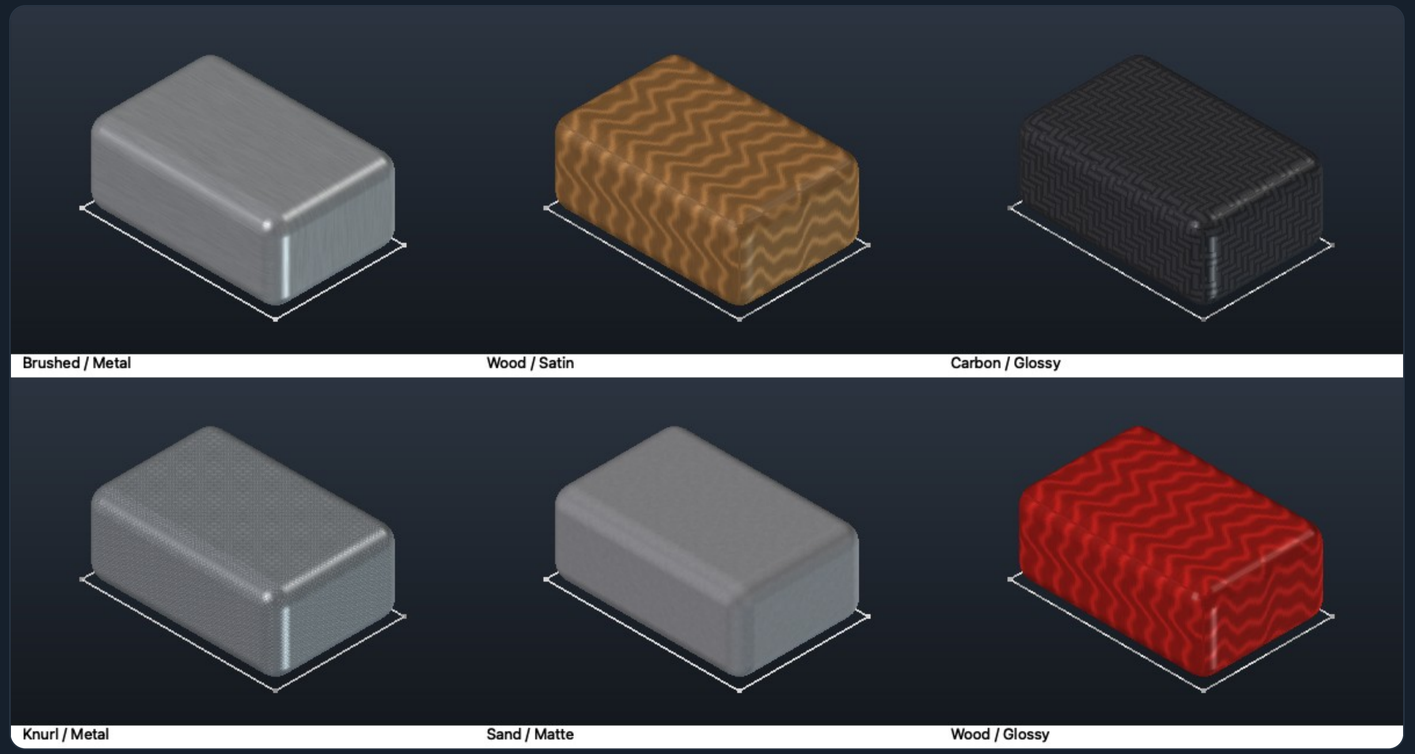




- Any colour: the palette or **More...**
- Finish: **Matte, Satin, Glossy, Metal, Chrome, Glass, Rubber**
- Colours the picked faces, Solids or the whole part; **Hide Edges** takes the edge lines off
- Shown at once, OK is one Undo step; a face keeps its colour when a feature moves it

APPEARANCE ▸ TEXTURE

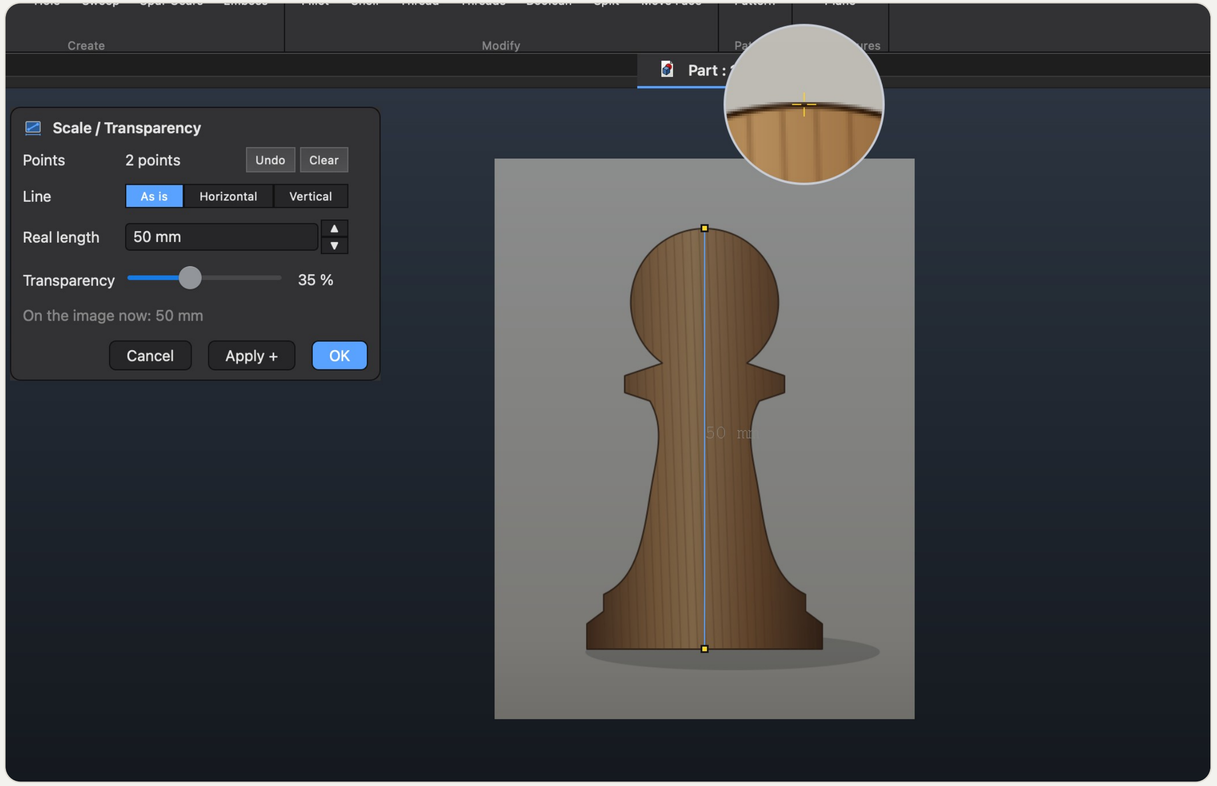
Textures over the colour

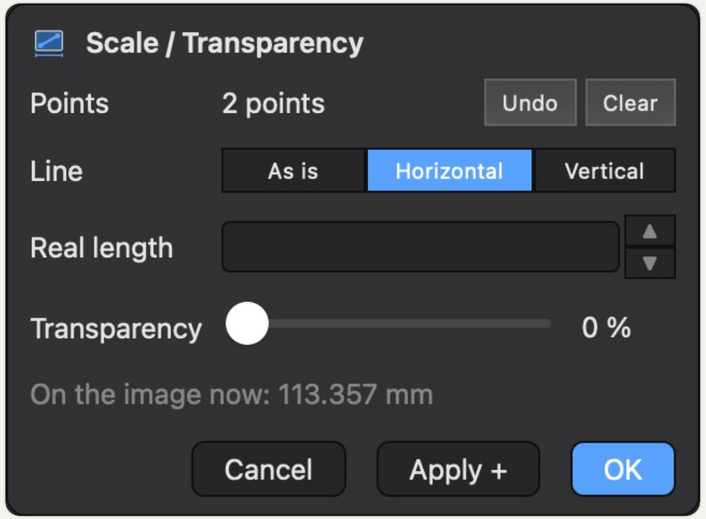


- **Brushed, Wood, Carbon, Knurl, Sand**
- The colour shows through: a red part gets red wood
- The pattern keeps its real size on any part
- Saved in the file, shown in assemblies too
- Photo rendering through Blender — some day; colour and textures for now

IMAGE TAB

A photo to sketch over, at its true scale

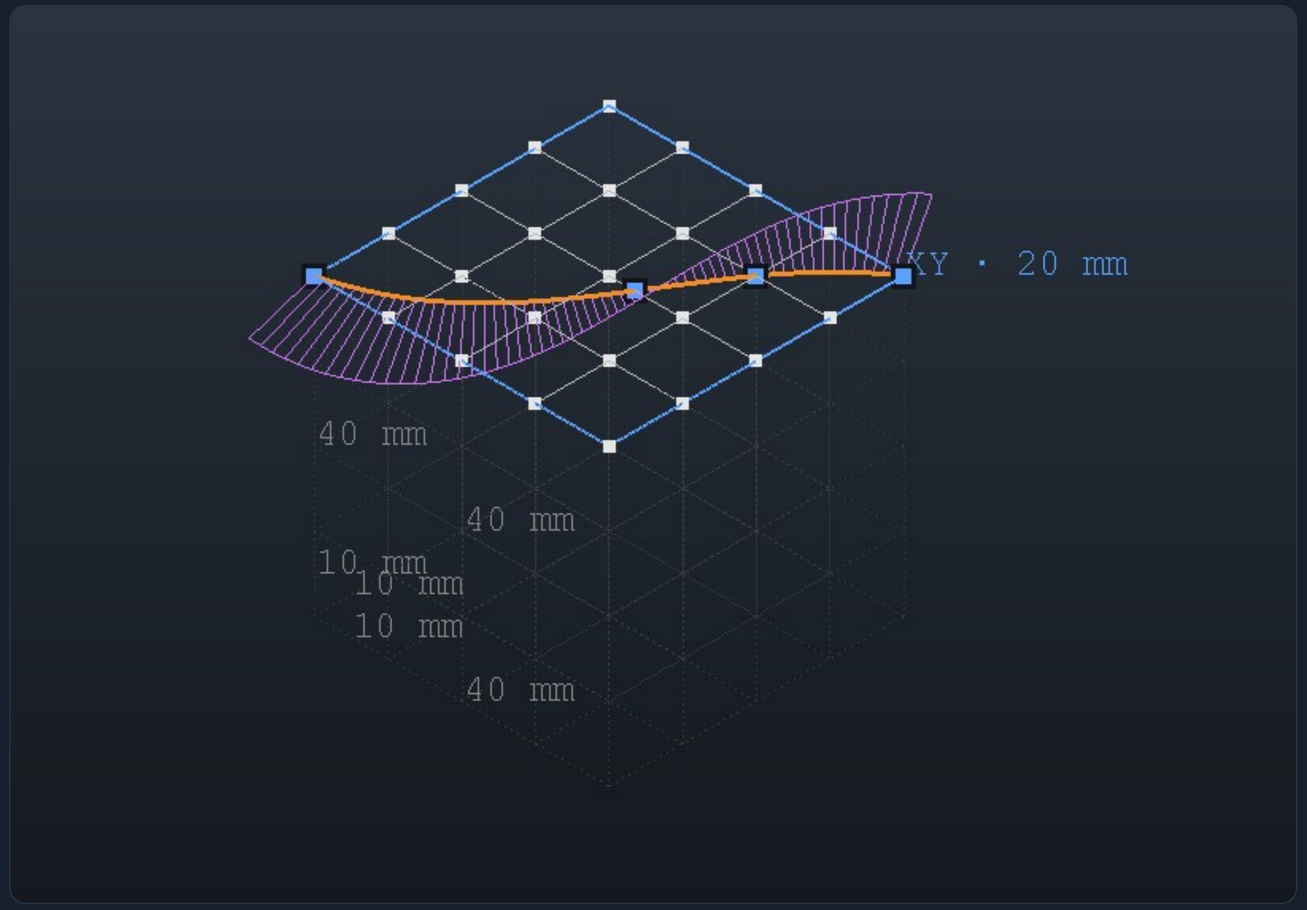


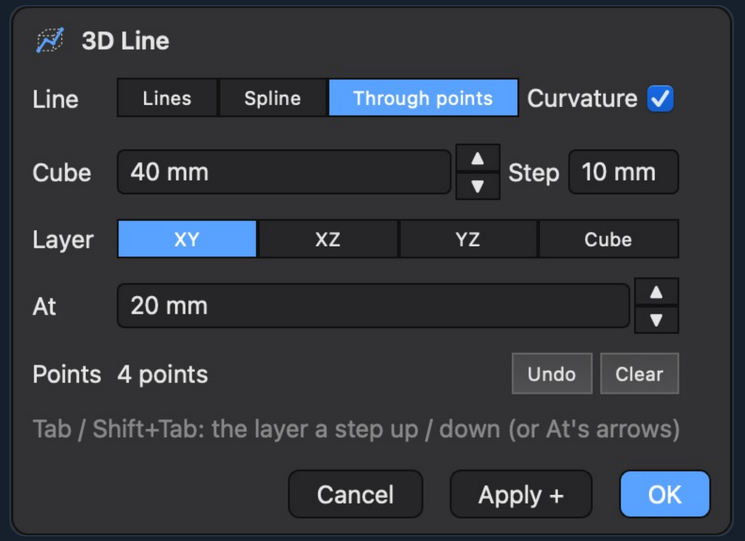


- **Insert Image**: JPG, PNG, BMP; On XY / XZ / YZ, Turn 90°, Mirror, Show / Hide
- **Scale**: two points and the real length — the whole picture scales
- Line ▸ **Horizontal / Vertical** turns the picture to level that line
- A magnifier under the pointer, small cross points; a transparency slider

3D MODEL ▸ SKETCH ▸ 3D LINE

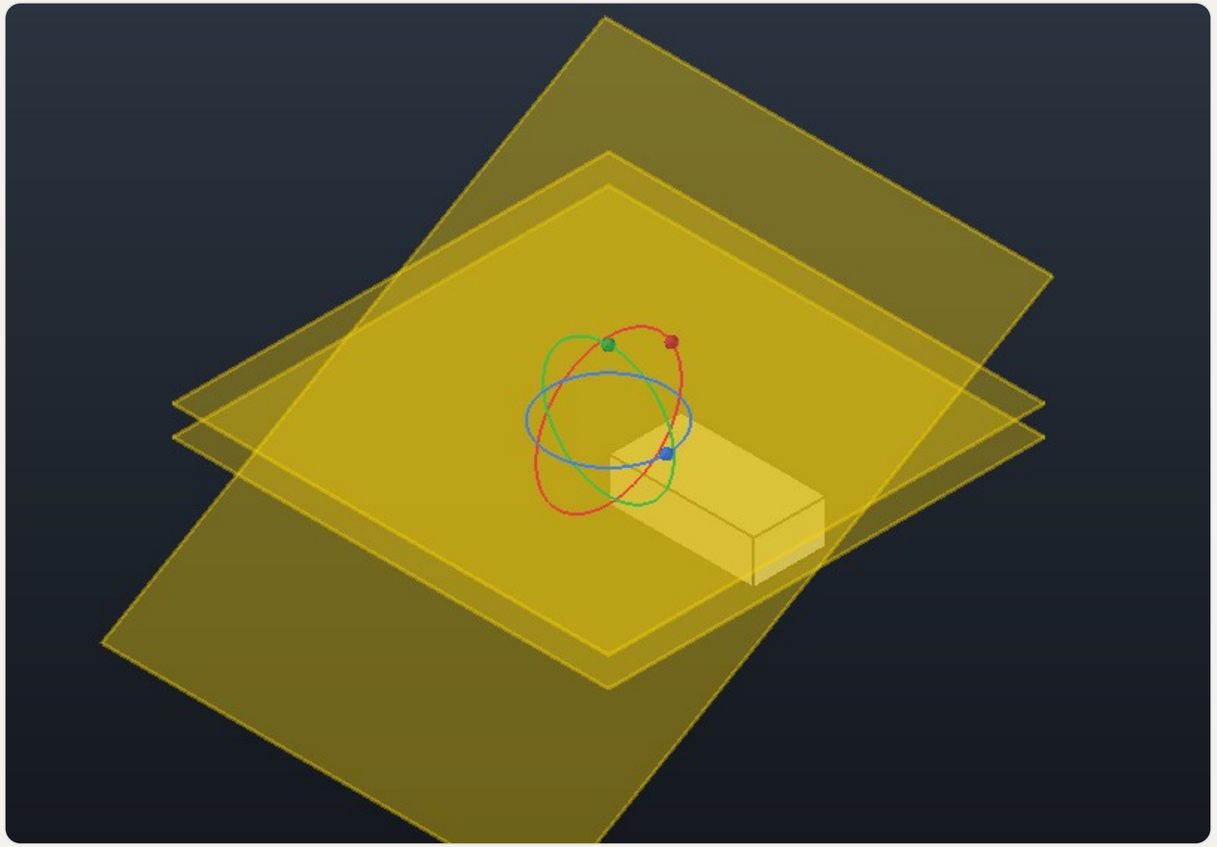
A line in space on a virtual cube

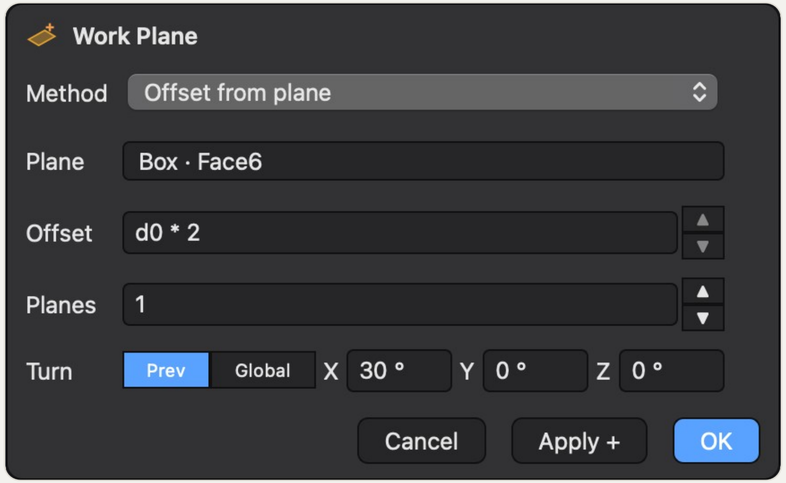




WORK PLANE

Loft planes — several at once, and turned

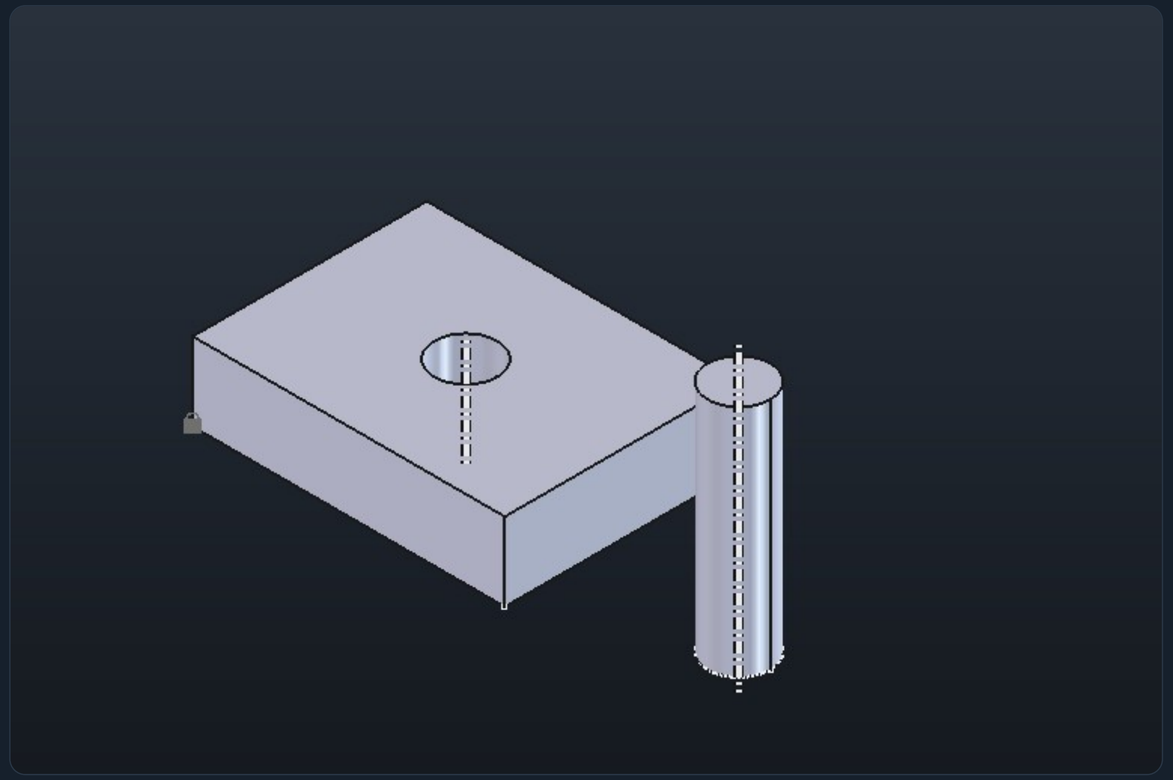




- Offset ▸ **Planes**: 2×, 3×… the offset in one go
- **Turn**: X, Y, Z about the previous plane or the global axes
- Typed or dragged on rings; the next plane stays as it was

ASSEMBLY

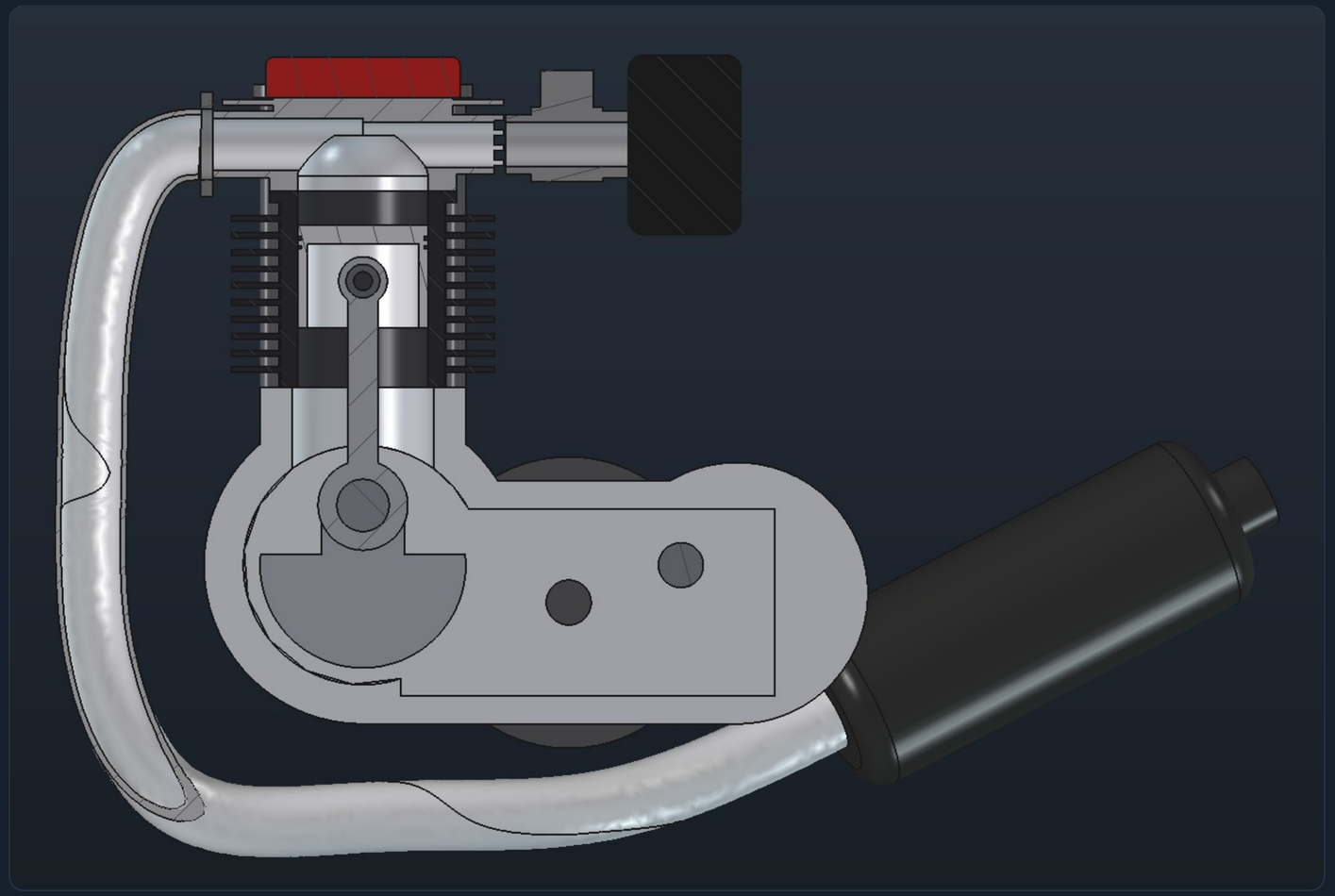
Axes to constrain to

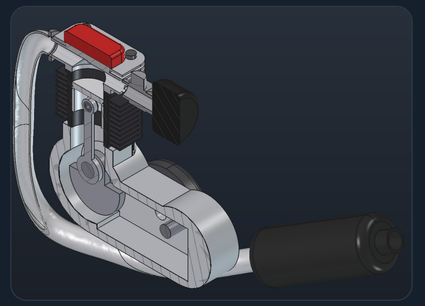


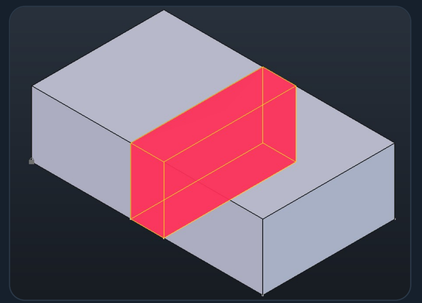
- Every turned body and hole shows its **axis** as a light centre line
- Mate, Flush, Insert, Cylindrical, Parallel… take an axis with a click
- The gizmo stays about 3 s after a drag — easy to grab again
- The grabbed handle **changes colour**; smaller arrows
- Make Assembly keeps the parts' names; the assembly is **name_assy**

ASSEMBLY · VIEW ▸ SECTION

The whole assembly cut — every part in its colour NEW IN 0.9





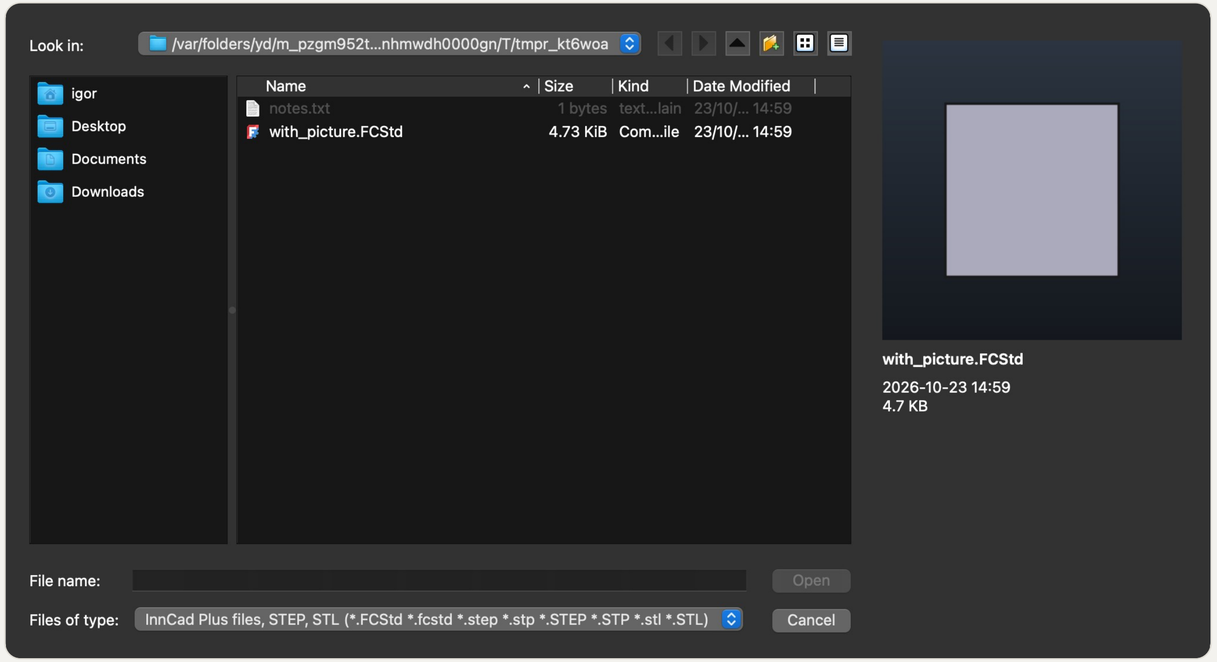


- The plane cuts every part it meets, each with its own hatching
- Any shape: InnCad and FreeCAD assemblies, STEP, other programs' files
- **Interference**: where parts go into each other, in bright red, with the volume
- Hide / Show for axes, sketches, origin; **Open** a part in its own window

Moto Engine Demo — a 20-part assembly made in FreeCAD

AND MORE

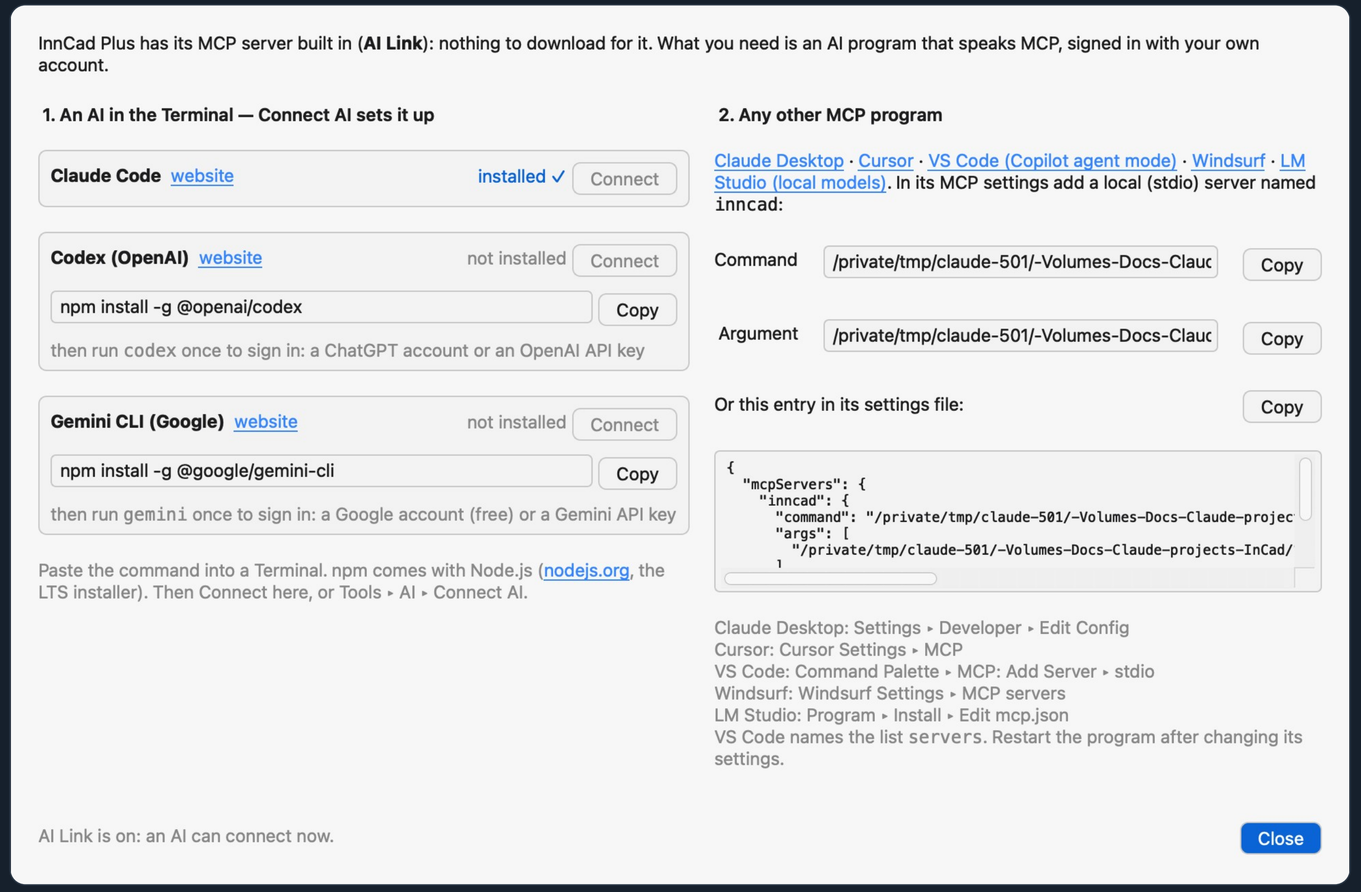
Small things that save time



- **Open** with previews: FCStd, STEP, STL — picture, date, size; only what opens is listed
- Import: **IGES**, **OBJ** (Plus), JPG / PNG to sketch over
- **Screenshot**: the 3D view or the whole window, PNG (transparent too), JPEG, BMP
- Extrude ⇄ 2: the second side has its own Distance / To Next / Through All / To
- Extrude ▸ To: a face, an edge, a point, an axis
- Fillet ▸ **Full**: a full round between two edges
- Tips in blue, errors in red

TOOLS ▸ AI ▸ CONNECT AI

One button, and your AI is in InnCad PLUS ONLY



- Claude Code installed? **Connect AI** opens a Terminal with it in the part's folder, no questions
- The MCP server is built into InnCad Plus — nothing to download
- Nothing installed? The window on the left: where to get Claude Code, Codex or Gemini CLI, and the install command
- Claude Desktop, Cursor, VS Code, Windsurf, LM Studio: ready paths and JSON with Copy

Try it!

Start with the pawn: photo → Scale → sketch → Revolve → Appearance. If something goes wrong, send the log (chapter 11 of the manual).

